PRD: ShieldZap — Privacy-Preserving AI DeFi Agent with One-Tap Yield

Date: 2026-03-07 **Domain:** AI x Crypto **Confidence Score:** 49 / 60 (+3 AI x Crypto bonus = 52) **Generated via:** /generate-prd **Target:** Starkzap Developer Challenge ($1,500 best build / $500 most creative)

Idea Scoring Matrix

Idea	Market Timing	Demand Signal	Competition Gap	Tech Feasibility	Monetization	Market Scope	Tot
1. ShieldZap — AI Agent + Private Yield	9	8	9	8	7	8	**49 (+3**
2. ZapPredict — Social Prediction Market	7	8	5	9	7	7	**43**
3. YieldPilot — Gamified Savings Dashboard	7	7	6	9	6	7	**42**
4. CopyZap — Social Copy Trading	6	7	4	8	7	8	**40**
5. TouchGrass — GPS Proof-of-Activity Staking	6	5	8	7	5	5	**36**

Idea Descriptions

1. ShieldZap — Privacy-Preserving AI DeFi Agent with One-Tap Yield An agentic app where users chat with an AI to privately stake BTC/STRK, earn yield, and manage shielded DeFi positions — all through natural language, with zero gas fees via Starkzap's paymaster and social login. Combines Obscura's ZK privacy engine with Starkzap's seamless onboarding.

- **Why now:** strkBTC launching Q1 2026, Starkzap SDK just released, AI agent wallets are the hottest narrative ($3B market), a16z flagged privacy + agents as top 2026 trends.

2. ZapPredict — Bet with Friends on Anything A social prediction market where friends create custom betting pools (sports, crypto prices, personal bets) using Starkzap's social login + gasless transactions. Payouts in BTC/STRK.

- **Why now:** Prediction markets at $6B/week, Polymarket $9B valuation. But no Starknet-native option with social login.

3. YieldPilot — Simplified One-Tap Crypto Yield A mobile-first yield dashboard that scans Starknet yield pools (Endur, Vesu, Nostra), compares APRs, and lets users enter with one tap. Gamified savings goals + streak rewards.

- **Why now:** Aave launching retail yield app on App Store. Gap on Starknet for simplified yield.

4. CopyZap — Social Copy Trading on Starknet Users follow and auto-copy trades of top Starknet DeFi performers. Uses Starkzap for social login + gasless execution.

- **Why now:** OKX just launched Orbit social network at $25B valuation. No decentralized copy trading on Starknet.

5. TouchGrass — Proof-of-Activity Staking Stake STRK/BTC, earn yield by sharing GPS location + photo of touching grass. Social leaderboard.

- **Why now:** Gamified DeFi trend + wellness culture. Creative but niche market.

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1. Executive Summary

ShieldZap is a privacy-preserving AI DeFi agent that lets users manage shielded Bitcoin and STRK positions through natural language chat, with zero-friction onboarding via Starkzap's social login and gasless transactions.

Core Problem: DeFi remains too complex for 95% of users. Privacy-conscious users face even worse UX — managing ZK proofs, encrypted balances, and multiple transactions. Meanwhile, AI agents are exploding but none offer privacy-preserving DeFi execution on Starknet.

Why NOW:

- Starkzap SDK launched Feb 24 with social login + paymaster (gasless) + DeFi modules
- strkBTC (private Bitcoin) launching Q1 2026 on Starknet mainnet
- AI agent wallets: Coinbase, OKX, MoonPay all launched agent infra in Feb-Mar 2026
- a16z flagged AI agents + privacy as the two dominant 2026 narratives
- starknet-agentic SDK provides agent account contracts with session keys
- $3.06B AI agent economy, 22.1% CAGR for ZK privacy market

Target Outcome (6 months): 5,000 users managing $2M+ in private DeFi positions through AI chat, with 50%+ retention as the go-to privacy DeFi interface on Starknet.

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2. Market Context & Research Backing

Key Trends

1. **AI Agent Economy ($3.06B)** — Coinbase, OKX, MoonPay, and Binance all launched agent wallet infrastructure in Q1 2026. x402 standard has processed 50M+ machine-to-machine transactions. (Source)

2. **Privacy as Institutional Requirement** — Starknet launching strkBTC with shielded mode for confidential DeFi. Goldman Sachs and Sony integrating ZKP. VP of Growth at Starknet Foundation: "Public blockchains expose sensitive positions." (Source)
3. **Starkzap Lowering Integration Friction** — SDK reduces crypto feature integration from "months to minutes." Social login, gasless transactions, DeFi modules out of the box. (Source)
4. **DeFi Yield Going Retail** — Aave launching yield app on Apple App Store with 5%+ APY. Yearn Finance automating yield optimization. (Source)

Market Size

- **TAM:** $50B blockchain AI market by 2030 + $7.59B ZK privacy market by 2033
- **SAM:** $3.06B AI agent economy (current) intersected with Starknet DeFi ($500M+ TVL)
- **SOM:** 5,000 privacy-conscious Starknet users × $400 avg position = $2M in year 1

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3. Problem Statement

Target Persona

Name: Alex, 28, crypto-native developer/trader **Context:** Holds BTC and STRK, wants yield but concerned about on-chain surveillance. Has tried Aave, Endur, but finds DeFi UX painful. Interested in privacy but can't navigate ZK proofs manually.

Current Pain

- DeFi requires 5-10 manual steps per operation (connect wallet, approve, stake, monitor)
- Privacy tools are even more complex (keygen, encryption, witness generation, proof submission)
- No single interface combines yield + privacy + simplicity
- Gas fees and wallet setup create friction for new users

Current Solutions

Solution	Cost/Time	Why It Fails
Manual DeFi (Vesu, Endur)	15+ min per operation	No privacy, complex UX
Railgun	Variable	No Starknet support, limited DeFi composability
Tornado Cash	High risk	Sanctioned, no yield integration
Yearn Finance	Gas fees	No privacy, Ethereum L1 only
Our own Obscura	30+ sec proof gen	Complex UI, no AI assistant for discovery

Gap

No product combines: (1) AI-driven natural language DeFi, (2) ZK privacy for balances and transactions, (3) zero-friction onboarding (social login + gasless), (4) yield optimization — all on Starknet.

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4. Proposed Solution

ShieldZap is a web app where users:

1. Sign in with Google/Apple/email (Starkzap social login — no seed phrase)
2. Chat with an AI agent: "Stake 0.1 BTC privately and earn yield"
3. The AI generates ZK proofs, encrypts balances, submits gasless transactions
4. Users see their private portfolio: shielded balances, yield earned, proof history

Core Value Proposition

"Private DeFi yield in one sentence — just tell the AI what you want."

Key Differentiators

1. **AI-first interface**: Natural language replaces complex DeFi UIs. "Stake 0.1 BTC" instead of 5 manual steps.
2. **Privacy by default**: ZK proofs (Noir circuits) + encrypted balances (ElGamal) — your portfolio is invisible on-chain.
3. **Zero-friction onboarding**: Starkzap social login + paymaster = no wallet setup, no gas tokens needed.
4. **Real proof verification**: Garaga on-chain verifiers — not mock verification, real ZK proofs verified on Starknet.
5. **Starknet-native**: Built for strkBTC + Starknet ecosystem, leveraging native account abstraction.

10x Better

- vs Manual DeFi: 1 sentence vs 10 steps, private vs public
- vs Existing privacy tools: AI handles proof complexity, social login vs seed phrases
- vs AI trading bots: Privacy-preserving, not just automated

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5. Key Features (MVP)

P0 — Must Ship

Feature	Description	Effort
Starkzap Social Login	Google/Apple/email sign-in via Starkzap Wallets module. No seed phrases.	2 days
AI Chat Agent	Natural language DeFi execution. "Stake 0.5 BTC", "What's my yield?", "Withdraw everything". Uses DeepSeek/Claude for intent parsing + action execution.	3 days
Gasless Transactions	All transactions sponsored via Starkzap Paymaster. User never buys gas.	1 day
Private Staking	Deposit BTC/STRK → stake via Endur → shield balance with ZK proof. Uses Obscura's proof pipeline.	3 days
Shielded Balance Dashboard	Locally-decrypted balance display. On-chain, balances are encrypted. Shows yield earned.	2 days
Proof History	Track all ZK proofs generated, verification status, circuit type, timing.	1 day

P1 — Should Ship

Feature	Description	Effort
Yield Comparison	Scan Starknet yield pools (Endur, Vesu, Nostra) and recommend best APR via AI	2 days
CDP Operations	Lock shielded collateral → mint sUSD through AI chat	2 days
Solvency Dashboard	Public solvency proofs showing vault/CDP health without revealing individual positions	1 day

P2 — Future

Feature	Description	Effort
strkBTC Integration	Native strkBTC shielded mode when mainnet launches	1 week
Multi-chain via LayerZero	Cross-chain private DeFi positions	2 weeks
Agent Autonomy	AI agent auto-rebalances yield positions with user-set guardrails (session keys)	1 week

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6. Technical Architecture

Tech Stack

Layer	Technology	Justification
Frontend	React 18 + TypeScript + Vite + TailwindCSS	Same as Obscura, fast iteration
Wallet	**Starkzap SDK** (Wallets + Paymaster modules)	Challenge requirement, best Starknet onboarding
AI	DeepSeek API (via Vercel serverless proxy)	Cost-effective, strong instruction following
ZK Proofs	Noir 1.0.0-beta.16 + bb.js (in-browser)	Proven in Obscura, Garaga compatible
On-chain Verification	Garaga verifier contracts (7 circuits)	Already deployed on Sepolia
Privacy	ElGamal encryption (Tongo pattern)	Proven in Obscura
Contracts	Cairo (Starknet) — ShieldedVault, ShieldedCDP	Already deployed on Sepolia
Hosting	Vercel	Free tier, serverless API routes

System Architecture (ASCII)

```
┌──────────────────────────────────────────────────────────┐
│                    USER (Browser)                        │
│                                                          │
│  ┌──────────┐  ┌──────────┐  ┌──────────────────┐       │
│  │ Starkzap │  │ AI Chat  │  │ Privacy Engine   │       │
│  │ Social   │  │ Widget   │  │ (ElGamal + Noir) │       │
│  │ Login    │  │          │  │                  │       │
│  └────┬─────┘  └────┬─────┘  └────────┬─────────┘       │
│       │             │                 │                 │
│  ┌────┴─────────────┴─────────────────┴──────────────┐  │
│  │          Action Executor (13 actions)             │  │
│  │  deposit | shield | stake | unshield | withdraw   │  │
│  │  open_cdp | lock | mint_susd | repay | close      │  │
│  └──────────────────────┬────────────────────────────┘  │
│                         │                               │
│  ┌──────────────────────┴────────────────────────────┐  │
│  │        Starkzap Paymaster (gasless tx)            │  │
│  └──────────────────────┬────────────────────────────┘  │
└────────────────────────┬┴───────────────────────────────┘
                         │
                         │
                ┌────────┴──────────┐
                │   Starknet (Sepolia)  │
                │                       │
                │  ShieldedVault        │
                │  ShieldedCDP          │
                │  ProofVerifier        │
                │  (Garaga verifiers)   │
                │  SolvencyProver       │
                │  MockPriceFeed        │
                └───────────────────────┘
```

Data Flow

1. User signs in via Starkzap (social login → Cartridge Controller wallet)
2. User types: "Stake 1 BTC privately"
3. AI parses intent → maps to actions: [faucet, deposit, shield]
4. For each action: a. Generate ElGamal encryption (client-side) b. Generate Noir ZK proof (in-browser via bb.js) c. Encode Garaga calldata d. Submit via Starkzap Paymaster (gasless)
5. On-chain: Garaga verifier validates proof → state updates
6. Frontend decrypts + displays updated shielded balance

Infrastructure Costs

- Vercel: Free tier (100GB bandwidth, serverless functions)
- DeepSeek API: ~$0.14/M input tokens, $0.28/M output = ~$5/month for 1K users
- Starknet Sepolia: Free (testnet)
- Total: **< $10/month for MVP**

Security Considerations

- Private keys never leave browser (Starkzap self-custody)

- ElGamal keys stored in encrypted browser storage
- ZK proofs generated client-side (witnesses never transmitted)
- AI system prompt sandboxed — can only execute predefined 13 actions
- XSS protection on AI-generated content (HTML entity escaping + sanitization)
- DeepSeek API key server-side only (Vercel proxy)

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7. User Flow

Primary Journey: "Stake BTC Privately"

```
Step 1: Land on shieldzap.app
        ┌──────────────────────────┐
        │   ShieldZap              │
        │                          │
        │   Private DeFi Yield     │
        │   powered by AI          │
        │                          │
        │   [Sign in with Google]  │
        │   [Sign in with Apple]   │
        │   [Connect Wallet]       │
        └──────────────────────────┘

Step 2: Starkzap social login (Google) → wallet created automatically

Step 3: AI Chat opens with greeting:
        "Hi! I'm your private DeFi agent. I can help you
         stake BTC/STRK, earn yield, and manage positions
         — all with zero-knowledge privacy. What would
         you like to do?"

Step 4: User types: "I want to stake 2 BTC and earn yield privately"

Step 5: AI confirms action:
        "I'll stake 2 BTC privately. Here's my plan:
         1. Get test BTC from faucet
         2. Deposit into ShieldedVault
         3. Generate ZK proof and shield your balance

         Your balance will be encrypted on-chain.
         Only you can see it. Execute?"
        [Execute] [Cancel]

Step 6: User clicks Execute → proof generation progress shown:
        ✓ Loading circuit...
        ✓ Generating witness...
        ● Generating ZK proof... (12s)
          Submitting to Starknet...

Step 7: Success! Dashboard updates:
        ┌──────────────────────────┐
        │  Shielded Balance        │
```

```
        │  ▇▇▇▇▇▇▇▇ 2.0 sxyBTC       │
        │  Yield: 5.2% APR           │
        │  Status: VERIFIED ✓        │
        │                            │
        │  Privacy Score: 95/100     │
        └────────────────────────────┘

Step 8: User asks: "What's my yield so far?"
        AI responds with decrypted balance + accumulated yield

Step 9: User asks: "Open a CDP with 1 sxyBTC and mint 500 sUSD"
        AI executes CDP flow with ZK proofs

Step 10: User asks: "Withdraw everything"
         AI executes unshield → withdraw with proofs
```

Onboarding (First 5 Minutes)

1. Sign in with Google (0 sec — Starkzap)
2. See AI greeting with examples (5 sec)
3. Type first command or click suggested action (10 sec)
4. Watch proof generation (15-30 sec)
5. See shielded balance on dashboard (immediate)

Core Loop (Daily Usage)

1. Open app → see shielded balance + yield
2. Ask AI: "How's my portfolio?" → AI shows decrypted stats
3. Optionally: rebalance, stake more, or withdraw via chat
4. Check proof history for verification status

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8. Monetization Strategy

Revenue Model

For the challenge: **Free / open-source** (maximize adoption + judge appeal). Post-challenge:

Tier	Price	Features
Free	$0	5 AI actions/day, basic yield
Pro	$9.99/mo	Unlimited actions, yield optimization, CDP
Agent	$29.99/mo	Auto-rebalancing AI agent, multi-position management

Pricing Benchmarks

- Aave retail app: Free (protocol fees ~0.1-0.5%)
- Yearn Finance: 2% management + 20% performance fee
- Railgun: Free (protocol level)
- AI trading bots (3Commas): $22-$49/month

Unit Economics (Post-Challenge)

- CAC: $15 (crypto Twitter + Starknet community)
- LTV: $120 (Pro tier × 12 months)
- Payback: 1.5 months

Revenue Projection

- Month 3: $2K MRR (200 Pro users)
- Month 6: $10K MRR (500 Pro + 100 Agent)
- Month 12: $50K MRR (2,000 Pro + 500 Agent)

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9. Go-to-Market Strategy

Week 1-2: Pre-Launch (Challenge Build Period)

- Build MVP using Starkzap SDK + Obscura contracts (already deployed)
- Create demo video showing AI chat → private staking flow
- Write Twitter thread: "What if DeFi was private AND simple?"
- Share progress in Starknet Discord

Launch Day (Challenge Submission)

- Submit PR to awesome-starkzap
- Post on Starknet Discord + Twitter
- Record 2-min demo video
- Publish README with architecture diagram

Month 1-3: Growth

- Starknet Foundation seed grant application ($25K)
- Integrate strkBTC when mainnet launches
- Content: "Why AI + Privacy = Future of DeFi" blog series
- Partner with Endur, Vesu for yield integration promotion

Month 3-6: Scaling

- Mobile app via Starkzap React Native SDK
- Mainnet deployment with real BTC
- Apply for Propulsion Program (gas rebates)
- Expand to multi-chain via LayerZero

Key Partnerships

1. **Starkzap** (SDK provider — tight integration showcase)
2. **Endur** (liquid staking — yield source)
3. **Vesu/Nostra** (lending — yield comparison)
4. **Tongo** (confidential ERC20 — privacy layer)

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10. Competitive Landscape

Direct Competitors

Feature	ShieldZap	Railgun	Aave App	Yearn	3Commas
AI Chat Interface	✅	❌	❌	❌	❌

ZK Privacy	✅	✅	❌	❌	❌
Social Login	✅	❌	❌	❌	❌
Gasless	✅	❌	❌	❌	N/A
Starknet Native	✅	❌	❌	❌	❌
Yield Optimization	✅	❌	✅	✅	✅
CDP/Borrowing	✅	❌	✅	❌	❌
On-chain Proofs	✅	✅	❌	❌	❌

Indirect Competitors

- **Coinbase Agentic Wallets**: Agent infra, not privacy-focused
- **OKX OnchainOS**: Multi-chain agent toolkit, no ZK privacy
- **MoonPay Agents**: Payment agents, not DeFi yield

Competitive Positioning

```
                    HIGH PRIVACY
                         |
               ShieldZap| Railgun
                   ●     |    ●
                         |
  SIMPLE ————————————————|———————————— COMPLEX
                         |
             Aave App ●  |  ● Yearn
                         |
                         |
                    LOW PRIVACY
```

SWOT Analysis

Strengths	Weaknesses
Only AI + privacy + Starkzap combo	New/unproven product
Contracts already deployed on Sepolia	In-browser proving is slow (15-30s)
7 real Garaga-verified circuits	Testnet only (no real funds)
Full proof pipeline working E2E	DeepSeek API dependency

Opportunities	Threats
strkBTC mainnet launch → first private BTC yield	Starkzap SDK bugs/breaking changes
Starknet Foundation grants ($25K-$1M)	Larger team builds similar product
AI agent narrative = media attention	Privacy regulation uncertainty
Challenge prize money → bootstrap	ZK proof UX still unfamiliar to users

Competitive Moat

1. **Technical depth**: 7 working Noir circuits with Garaga on-chain verification — takes months to replicate
2. **First-mover on Starknet privacy + AI**: No competitor combines both
3. **Switching costs**: Encrypted state is protocol-specific, users can't easily migrate private positions
4. **Composability**: CDP on top of shielded vault creates lock-in through utility

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11. Risk Assessment

Risk	Severity	Likelihood	Mitigation
Starkzap SDK instability	High	Medium	Pin SDK version, abstract wallet layer, report issues to earn challenge bonus
In-browser proof too slow	Medium	High	Show progress UI, pre-load circuits, consider proof delegation in v2
DeepSeek API rate limits	Medium	Low	Cache common responses, fallback to local intent parsing
Challenge judges favor simpler apps	Medium	Medium	Polish UX, create compelling demo video, emphasize "most creative" angle
Privacy regulations tighten	High	Low	KYA-ready architecture, compliance-optional shielded mode
Users don't understand privacy value	Medium	Medium	Education content, show "what attackers see" vs "what you see" comparison

Kill Criteria

- If < 50 users try the app in first month post-launch → pivot to simpler yield aggregator
- If Starkzap SDK doesn't support paymaster integration → fall back to standard wallet connect
- If proof generation consistently fails in browsers → pivot to server-side proving
- If strkBTC doesn't launch → focus on STRK staking privacy instead

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12. Success Metrics & Validation

Primary North Star Metric

Private DeFi positions created via AI chat (measures core value delivery)

Input Metrics (Leading)

1. AI chat messages sent per session
2. Proof generation success rate
3. Social login completion rate (Starkzap)
4. Time from sign-in to first private position

Output Metrics (Lagging)

1. Total shielded value (TVL)
2. Weekly active users

3. CDP positions opened
4. User retention (7-day, 30-day)
5. Challenge score / prize placement

Targets

Metric	30 days	60 days	90 days
Users	100	500	2,000
Private positions	50	300	1,500
TVL (testnet)	$10K equiv	$100K equiv	$500K equiv
Proof success rate	85%	92%	97%
Retention (7-day)	25%	35%	45%

Validation Experiments

Experiment 1: "Does AI simplify DeFi?"

- Test: Give 10 non-crypto users the app. Can they stake BTC in < 2 minutes?
- Success: 7/10 complete first stake via AI chat without help

Experiment 2: "Do users value privacy?"

- Test: A/B test landing page — "AI DeFi Agent" vs "Private AI DeFi Agent"
- Success: Privacy version gets 30%+ higher sign-up rate

Experiment 3: "Is Starkzap onboarding frictionless?"

- Test: Measure drop-off from landing page → first action
- Success: < 20% drop-off (vs industry avg 60-70% for wallet-connect flows)

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13. MVP Development Timeline

Phase 1 (Day 1-3): Foundation

- Integrate Starkzap SDK (social login + paymaster)
- Connect to existing Obscura contracts on Sepolia
- Basic app shell with auth flow
- **Deliverable:** User can sign in with Google and see wallet address

Phase 2 (Day 4-6): AI Agent

- Port AIChat component with Starkzap wallet context
- Wire executor to use Starkzap paymaster for gasless tx
- DeepSeek system prompt with Starkzap-specific actions
- **Deliverable:** User can chat with AI and execute deposit/shield

Phase 3 (Day 7-9): Privacy Pipeline

- Connect Noir proof generation to Starkzap-managed wallet
- Shielded balance display with local decryption
- Proof history with verification status
- **Deliverable:** Full stake-shield-unshield flow via AI chat

Phase 4 (Day 10-12): Polish & Submit

- Landing page with value proposition
- Demo video recording
- README documentation
- Bug fixes, edge cases
- **Deliverable:** Challenge submission (PR to awesome-starkzap)

Team Requirements

- 1 full-stack developer (existing Obscura expertise): 12 days, 8hr/day
- Total: ~96 hours of development

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14. CEO/CTO Decision Brief

Build Confidence: **HIGH**

Reasoning:

- 80% of the backend is already built (Obscura contracts, 7 circuits, Garaga verifiers — all deployed on Sepolia)
- Frontend proof pipeline is working E2E
- Starkzap integration is primarily a wallet/paymaster swap (days, not weeks)
- AI chat assistant already functional with 13 action types

Key Assumptions

1. Starkzap SDK social login works smoothly with Cartridge Controller
2. Starkzap Paymaster can sponsor our contract calls
3. In-browser Noir proof generation remains functional with Starkzap's wallet abstraction
4. Challenge judges value technical depth (real ZK proofs) over simple UI demos

Biggest Unknown

Starkzap Paymaster compatibility with complex multi-call transactions (proof verification + state mutation in one tx). Mitigation: Test paymaster integration on day 1; fall back to user-paid gas if needed.

If We Had Only 2 Weeks

Build exactly the MVP above: Starkzap social login → AI chat → private BTC staking → shielded balance display. Skip CDP, yield comparison, and advanced features.

Comparable Exits/Acquisitions

- **Railgun**: Privacy protocol, ~$80M FDV
- **Aztec**: Privacy L2, raised $100M (Series B, Dec 2022)
- **3Commas** (trading bots): $1B+ valuation (2021)
- **Aave**: $2.5B FDV, now launching retail yield app
- **AI agent infrastructure**: Fetch.ai ($1.8B FDV), Autonolas ($500M FDV)

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15. Appendix: Research Sources

Starkzap & Starknet

- Starkzap: Bitcoin in your app

- Starkzap GitHub
- Starknet-Agentic Specification
- strkBTC: Shielded Bitcoin
- Starknet DeFi Ecosystem
- Starknet January 2026 Recap
- StarkWare on Starkzap

AI x Crypto

- a16z 2026 Predictions: AI Agents, KYA, Privacy
- Coinbase Agentic Wallets
- OKX OnchainOS for AI Agents
- MoonPay Agents Launch
- AI Agent Economics 2026
- Electric Capital: Agent Wallets Legal Frontier

ZK Privacy

- ZKP Solving AI Privacy 2026
- ZK Proof AI Secure Verifiable
- Top 5 ZKP Use Cases in DeFi 2026

Markets & Competition

- Prediction Markets Crypto Guide
- OKX Building Social Network (Orbit)
- Aave Retail Yield App
- Best Yield Farming Platforms 2026
- AI & Crypto Top 10 Trends 2026